

# Training-free, physics-verified LLM recipe generation for industrial machinery condition monitoring

Author One<sup>a</sup>, Author Two<sup>a</sup>

<sup>a</sup>Department, City, Country

---

## Abstract

Training a signal generator for each machine and data regime can be impractical when fault examples are scarce. This paper studies a different design point: an LLM proposes a structured signal recipe, a deterministic renderer produces the waveform, and train-calibrated physical gates decide whether the candidate may enter an augmentation pool. The gates are admission records, not proofs of physical validity. Under a file-disjoint PU bearing protocol, the frozen Phase-A v2 experiment shows that LLM recipes outperform rule recipes and random open-loop recipes at  $n_{\text{real}} = 5, 10, 25$  for both Accuracy and Macro-F1 after the registered Holm correction (20 seeds). A separate CWRU within-load and leave-one-load-out protocol yields 90/90 registered LLM-comparator tests passing Holm correction (40 seeds). On Berkeley milling, the healthy-versus-degraded binary formal test is deliberately reported as partial: 15/18 registered comparisons pass, including all 12 comparisons against non-structured baselines; against the rule baseline, only Accuracy at  $n_{\text{real}} = 2$  and both metrics at  $n_{\text{real}} = 5$  pass, while both metrics at  $n_{\text{real}} = 10$  do not. Physical diagnostics are reported as reference-relative evidence rather than a global generator ranking. We also report six failed PU cross-condition attempts, experimental-condition and process-metadata confounding in UMich milling, and a stopped MU-TCM inner validation line. These results support an auditable, training-free recipe admission workflow within the stated few-shot protocols, while defining its current boundary for cross-condition extrapolation and confounded milling data.

**Keywords:** Condition monitoring, Large language model, Synthetic data, Physics-based verification, Few-shot diagnosis, Training-free generation

---

## 1. Introduction

Synthetic augmentation is often used when labelled fault data are scarce, but the usual solution is to train a GAN, VAE, or diffusion model on the available signals. The cost and data requirement of that training are a poor fit for a small new dataset, a changing machine configuration, or a closed-source model that cannot be updated. LLMs offer another route: they can propose a compact, machine-conditioned recipe that a deterministic renderer turns into a signal window. The resulting flexibility does not make the output trustworthy. A plausible recipe can still yield a waveform with unsupported statistics, spectral structure, or fault-frequency evidence.

BREEZE treats this problem as admission rather than waveform repair. The LLM proposes a recipe; the renderer has responsibility for the signal equation; and a verifier, calibrated only on the training split, admits or rejects the rendered candidate. Failed candidates receive a structured report for a bounded number of resampling attempts. The verifier is not trained, it does not inspect the test split, and it does not modify a rejected waveform. A pass is therefore an auditable statement that a candidate met the configured, train-supported checks, not a certificate of physical correctness or downstream generalization.

This distinction matters for evaluation. Window-level random splits can place closely related windows from the same recording on both sides of a split, so their accuracy should not be compared directly with results from file- or experiment-disjoint protocols. We use only the frozen split manifests and paired statistical families registered for each dataset. We also keep negative experiments in the paper: six PU cross-condition attempts failed, and two milling lines were stopped before a formal test. These are boundaries of the method, not missing positive results.

The contribution is fourfold. First, we present a training-free recipe-renderer-verifier workflow for condition-monitoring signals, with clear responsibility boundaries. Second, we evaluate it under strict few-shot protocols on PU and CWRU, with paired Holm-corrected comparisons. Third, we report a qualified Berkeley milling result and class-conditional physical diagnostics rather than claiming universal or state-of-the-art superiority. Fourth, we document where the evidence fails: cross-condition morphology does not extrapolate under the tested PU protocols, and available milling metadata can confound wear labels.

## 2. Related work and positioning

GANs, VAEs, and diffusion models learn a signal distribution through model training [1, 2, 3, 4]. Recent bearing systems also combine language-derived descriptions with a learned diffusion generator, as in BearGen [5]. Such systems solve a different problem from BREEZE: they invest data and optimization into a learned generator. BREEZE does not claim to replace them. Its question is whether a candidate proposed by a black-box LLM recipe can be rendered and screened without training a new generator for the target machine.

This is also different from direct code or waveform generation. In a recipe-as-data design, the LLM produces named physical parameters and a deterministic renderer executes the signal model. The separation makes it possible to audit the proposal, renderer, and verifier independently. Bearing diagnostic tools such as envelope analysis and spectral kurtosis provide the domain evidence used by the verifier [6, 7, 8]; they do not turn the verifier into a complete physical simulator. Current sideband evidence is used only when the training data support it reliably [9].

## 3. Method

Let  $\mathcal{D}_{\text{tr}} = \{(x_i, y_i, m_i)\}$  denote training windows, labels, and available machine metadata. For class  $y$  and metadata  $m$ , an LLM proposes a recipe  $r_t$  at feedback round  $t$ . A deterministic renderer  $R$  constructs  $\tilde{x}_t = R(r_t)$ , and a verifier  $V$

returns a pass indicator and an auditable report:

$$(a_t, q_t) = V(\tilde{x}_t, y, m, \mathcal{D}_{\text{tr}}), \quad a_t \in \{0, 1\}. \quad (1)$$

The candidate is admitted only when  $a_t = 1$ . Otherwise, the next prompt receives the structured report  $q_t$  until the predeclared limit  $K$  is exhausted. No test window enters  $R$ ,  $V$ , or feedback construction.

The active gates check finite format and non-degeneracy, train-supported time-domain statistics, spectral shape, class-relevant envelope evidence, reliability-gated current evidence where applicable, and diversity relative to the accepted pool. Each check is calibrated on the appropriate outer training fold. The gate family is intentionally mixed: some features require an interval, some impose an upper or lower bound, and diversity is a nearest-neighbour exclusion rule. This is an engineering admission criterion, not a likelihood model or a proof that a window belongs to the real data distribution.

#### 4. Experimental protocols

*PU Phase-A v2.* The primary protocol uses the PU N09\_M07\_F10 file split. It contains 3846 training windows and 962 test windows over healthy, outer-race (OR), and inner-race (IR) classes. The renderer and verifier use only the training partition. The frozen comparison uses balanced pools of 150 synthetic windows per class, 20 paired CNN seeds,  $n_{\text{real}} = 5, 10, 25$ , and registered one-sided Wilcoxon/Holm families for LLM versus rule and random open-loop recipes.

*CWRU.* The frozen CWRU patch-v2 protocol uses 40 seeds,  $n_{\text{real}} = 5, 10, 25$ , and  $n_{\text{syn}} = 20$  per class. It evaluates a within-load0 split and four leave-one-load-out folds. The CWRU verifier has its own bearing configuration; no PU current-sideband gate is reused by default.

*Berkeley milling.* The formal milling task is binary: healthy means  $VB < 0.2$  and degraded means  $VB \geq 0.2$ . It uses a one-time 40-seed formal test at  $n_{\text{real}} = 2, 5, 10$  with  $n_{\text{syn}} = 20$  per class. The registered comparison family contains LLM versus rule, noise augmentation, and random open-loop for Accuracy and Macro-F1. Its result is reported with the full pass/fail pattern, rather than reduced to a single aggregate claim.

*Training-based baselines.* TimeGAN and a 1D DDPM are evaluated under the frozen PU file split using the same downstream CNN and shot counts. Each generator is trained only on the outer training data, once in a full-training-fold regime and once in a few-shot regime. The implementation checkpoints every epoch and records raw optimization trajectories, failures, generated-pool provenance, and wall time. These results are not used in the claims below until the predeclared 40-seed runs complete; smoke outputs are explicitly excluded.

Table 1: PU Phase-A v2 file-split results (20 seeds;  $n_{\text{syn}} = 150$  per class). Holm  $q$  values test LLM  $K = 3$  superiority within each registered family.

| $n_{\text{real}}$ | metric   | LLM $K = 3$       | rule              | random            | $q_{\text{rule}}$ | $q_{\text{random}}$ |
|-------------------|----------|-------------------|-------------------|-------------------|-------------------|---------------------|
| 5                 | Acc.     | $0.777 \pm 0.027$ | $0.696 \pm 0.053$ | $0.450 \pm 0.029$ | <0.001            | <0.001              |
| 5                 | Macro-F1 | $0.776 \pm 0.026$ | $0.691 \pm 0.049$ | $0.436 \pm 0.053$ | <0.001            | <0.001              |
| 10                | Acc.     | $0.796 \pm 0.030$ | $0.724 \pm 0.042$ | $0.483 \pm 0.013$ | <0.001            | <0.001              |
| 10                | Macro-F1 | $0.800 \pm 0.029$ | $0.724 \pm 0.040$ | $0.488 \pm 0.015$ | <0.001            | <0.001              |
| 25                | Acc.     | $0.818 \pm 0.021$ | $0.769 \pm 0.025$ | $0.529 \pm 0.022$ | <0.001            | <0.001              |
| 25                | Macro-F1 | $0.821 \pm 0.022$ | $0.770 \pm 0.024$ | $0.527 \pm 0.017$ | <0.001            | <0.001              |

Table 2: CWRU within-load0 results (40 seeds). All 90/90 registered LLM-comparator tests across within and four LOLO folds pass Holm correction; the table gives the within-load0 means.

| $n_{\text{real}}$ | metric   | LLM               | rule              | noise augmentation |
|-------------------|----------|-------------------|-------------------|--------------------|
| 5                 | Acc.     | $0.564 \pm 0.068$ | $0.549 \pm 0.053$ | $0.454 \pm 0.066$  |
| 5                 | Macro-F1 | $0.567 \pm 0.075$ | $0.554 \pm 0.062$ | $0.460 \pm 0.084$  |
| 10                | Acc.     | $0.621 \pm 0.037$ | $0.584 \pm 0.033$ | $0.534 \pm 0.065$  |
| 10                | Macro-F1 | $0.646 \pm 0.050$ | $0.611 \pm 0.046$ | $0.554 \pm 0.082$  |
| 25                | Acc.     | $0.674 \pm 0.037$ | $0.652 \pm 0.037$ | $0.628 \pm 0.055$  |
| 25                | Macro-F1 | $0.711 \pm 0.030$ | $0.689 \pm 0.035$ | $0.655 \pm 0.053$  |

## 5. Results

### 5.1. PU primary condition

The primary result is confined to the frozen PU file-split protocol. At all three registered few-shot counts, LLM  $K = 3$  recipes are significantly better than both rule and random-open-loop recipes for Accuracy and Macro-F1 after within-family Holm correction (Table 1). The result does not establish zero-real-data operation: the verifier and recipe construction depend on the outer training fold and class exemplars.

### 5.2. CWRU within-load and leave-one-load-out

On CWRU, the frozen within-load means favour LLM recipes over the rule and noise baselines at each shown shot count (Table 2). The stronger statistical statement is the registered one: all 90 LLM-comparator tests across the within-load and four LOLO folds pass their Holm correction. This result is specific to the CWRU split manifests and its train-calibrated verifier; it is not evidence that a single bearing configuration transfers without recalibration.

### 5.3. Berkeley milling is a qualified, partial result

Berkeley is not reported as a passed milling benchmark. The formal result has 15/18 passing registered comparisons. All 12 LLM comparisons against the non-structured noise and random-open-loop baselines pass. The rule

Table 3: Berkeley binary milling formal test (40 seeds;  $n_{\text{syn}} = 20$  per class). This is a partial result: all comparisons against noise augmentation and random open-loop pass, whereas LLM versus rule passes only Acc. at  $n_{\text{real}} = 2$  and both metrics at  $n_{\text{real}} = 5$ .

| $n_{\text{real}}$ | metric   | LLM               | rule              | noise aug.        | random open-loop  |
|-------------------|----------|-------------------|-------------------|-------------------|-------------------|
| 2                 | Acc.     | $0.786 \pm 0.021$ | $0.783 \pm 0.022$ | $0.695 \pm 0.110$ | $0.524 \pm 0.147$ |
| 2                 | Macro-F1 | $0.778 \pm 0.018$ | $0.776 \pm 0.019$ | $0.675 \pm 0.112$ | $0.450 \pm 0.127$ |
| 5                 | Acc.     | $0.784 \pm 0.030$ | $0.779 \pm 0.031$ | $0.724 \pm 0.068$ | $0.614 \pm 0.097$ |
| 5                 | Macro-F1 | $0.773 \pm 0.031$ | $0.769 \pm 0.032$ | $0.709 \pm 0.071$ | $0.514 \pm 0.106$ |
| 10                | Acc.     | $0.787 \pm 0.024$ | $0.786 \pm 0.024$ | $0.750 \pm 0.049$ | $0.654 \pm 0.062$ |
| 10                | Macro-F1 | $0.777 \pm 0.024$ | $0.776 \pm 0.023$ | $0.734 \pm 0.056$ | $0.572 \pm 0.081$ |

Table 4: Class-averaged PU physical diagnostics for fixed 150-window/class pools. Distances and errors are reference-relative diagnostics, not a downstream ranking; diversity has no preferred direction.

| pool             | RMS W1 | PSD W1 | band-energy error | NN diversity |
|------------------|--------|--------|-------------------|--------------|
| llm              | 0.008  | 236.9  | 0.044             | 17.10        |
| rule             | 0.031  | 271.6  | 0.235             | 14.20        |
| random_open_loop | 0.024  | 516.1  | 0.749             | 52.50        |
| noise_aug        | 0.004  | 309.1  | 0.014             | 46.09        |

comparison is narrower: at two shots, only Accuracy passes; at five shots, both metrics pass; and at ten shots, neither metric passes. Thus the supported claim is that structured physical recipes outperform all non-structured baselines in this experiment, while LLM recipes have an advantage over the rule baseline only in the lower-shot settings.

#### 5.4. Physical diagnostics

Physical diagnostics compare class-conditional synthetic pools with the matching outer-training reference. They include fault-frequency envelope alignment for bearing faults, RMS and kurtosis Wasserstein distances, PSD-CDF distance, band-energy error, and nearest-neighbour diversity. The main-text PU summary in Table 4 is generated from fixed 150-window/class pools; CWRU and Berkeley tables, complete class-wise values, source hashes, and availability records are retained with the experiment artifacts.

These diagnostics are not a universal ranking. For example, a single fault-frequency alignment diagnostic can favour a rule pool even when other reference-relative diagnostics and the downstream result differ. A matching frozen CWRU random-open-loop pool does not exist, and the table records that absence instead of substituting a different pool. TimeGAN and DDPM pools will be added only after their non-smoke formal runs finish.

## 6. Negative results and scope boundaries

The scope boundaries are part of the evidence. Six preregistered PU LOCO attempts failed. The sequence progressed from frequency mismatch through insufficient frequency correction and morphology extrapolation failures to a source-only CSCoh diagnostic that was non-discriminative for the target classes. We therefore do not claim cross-speed or cross-load morphology transfer.

For UMich milling, the correct conclusion is not that the signals are generically unlearnable. The available experimental-condition and process metadata are confounded with wear labels under the current split, and the confounding cannot be disentangled from wear without changing the protocol or acquiring additional metadata. For MU-TCM v3, the inner-validation gate passes only 2/6 core comparisons. No preregistration or formal held-out test was run. Neither line is used as a positive result.

## 7. Discussion

The results support a limited practical statement. A recipe-based, training-free admission workflow can improve few-shot diagnosis under the specified PU and CWRU protocols, and its decisions can be inspected through renderer inputs and gate reports. It is not a substitute for a learned generator in every data regime. The current trained-baseline study is designed to quantify that trade-off directly through matched downstream performance, data requirement, and measured training cost; no cost superiority claim is made before that study completes.

The Berkeley result illustrates why statistical detail matters. An average accuracy alone would hide the fact that rule and LLM converge at ten shots, and that the aggregate result is only partial. Similarly, physical fidelity is not synonymous with downstream utility. The verifier gives an auditable admission decision, but a candidate can satisfy the measured gates without capturing every mechanism relevant to a classifier or to a field deployment.

## 8. Conclusion

BREEZE separates LLM recipe proposal, deterministic signal rendering, and train-calibrated physical admission. On frozen file- and load-disjoint protocols, it yields positive few-shot results on PU and CWRU, a qualified Berkeley result, and explicit negative evidence for cross-condition and confounded milling settings. The central contribution is therefore an auditable workflow with stated limits, not a claim of universal generator superiority or zero-data fault diagnosis.

### Reproducibility and data availability

All numerical tables in this manuscript are generated from frozen CSV files by the script at [scripts/build\\_paper\\_tables.py](#). The source ledger is `analysis/evidence_ledger.md`; physical-metric provenance and availability records are in `analysis/section_C_physics_freeze_report.md`. The PU, CWRU, and Berkeley result

directories preserve their respective split manifests and registered test families. The Paderborn and CWRU data are public benchmark data. Berkeley milling preprocessing, aggregate outputs, and code can be shared subject to the originating data-use terms.

### **Declaration of Generative AI and AI-Assisted Technologies**

During preparation of this work, the authors used OpenAI Codex to assist with code inspection, experiment scripting, result-table assembly, and manuscript drafting. The authors reviewed and verified the content and take responsibility for the final manuscript.

### **References**

- [1] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial nets, in: *Advances in Neural Information Processing Systems*, 2014.
- [2] D. P. Kingma, M. Welling, Auto-encoding variational Bayes, in: *International Conference on Learning Representations*, 2014.
- [3] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, in: *Advances in Neural Information Processing Systems*, 2020.
- [4] J. Yoon, D. Jarrett, M. van der Schaar, Time-series generative adversarial networks, in: *Advances in Neural Information Processing Systems*, 2019.
- [5] J. Lee, H. Jeon, U. Kim, M. Kim, BearGen: LLM-guided signal generation framework for bearing fault diagnosis, *Advanced Engineering Informatics* 71 (2026) 104400. <https://doi.org/10.1016/j.aei.2026.104400>.
- [6] R. B. Randall, J. Antoni, Rolling element bearing diagnostics—a tutorial, *Mechanical Systems and Signal Processing* 25 (2011) 485–520.
- [7] J. Antoni, The spectral kurtosis: a useful tool for characterising non-stationary signals, *Mechanical Systems and Signal Processing* 20 (2006) 308–331.
- [8] J. Antoni, Fast computation of the kurtogram for the detection of transient faults, *Mechanical Systems and Signal Processing* 21 (2007) 108–124.
- [9] R. R. Schoen, T. G. Habetler, F. Kamran, R. G. Bartheld, Motor bearing damage detection using stator current monitoring, *IEEE Transactions on Industry Applications* 31 (1995) 127–138.